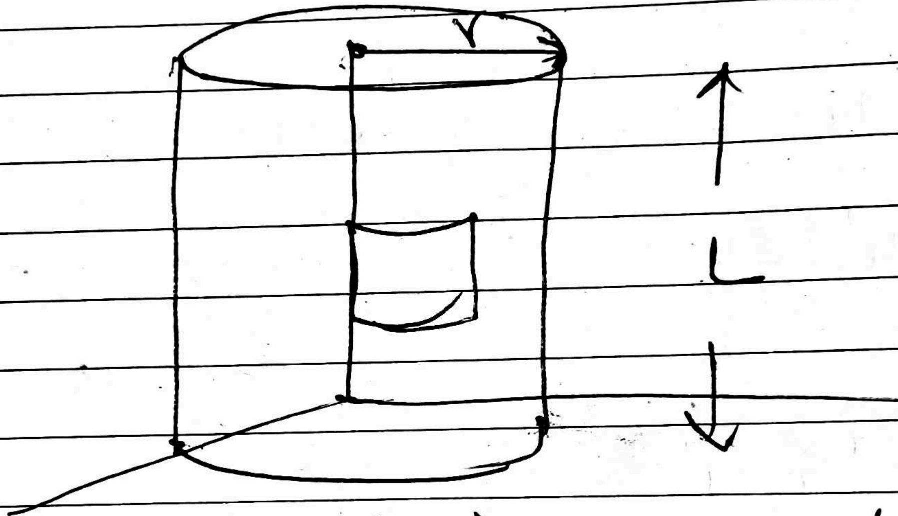
Question 3

(a) $\oint \vec{D} \cdot ds = \cancel{Q_{enclosed}}$

Taking an arbitrary cylindrical Gaussian surface of radius r over the region of space.



Therefore now applying the integral form of the Gauss law

$$\int_s \vec{D} \cdot d\vec{s} = Q_{enclosed}$$

$$d\vec{s} = r\,d\phi\,dz\,\hat{r}$$

~~Assum~~

Since the charge distribution is symmetric

Therefore

$$\vec{D} = D_r\,\hat{r}$$

$\therefore$

$$\int_c \vec{D}\cdot d\vec{s} = \int_c D_r \hat{r} \cdot r\,d\phi\,dz\,\hat{r}$$

$$\cancel{= D_r \int_s}$$

$$= r D_r \int_s d\phi\,dz$$

$$= r D_r \int_0^L \int_0^{2\pi} d\phi\,dz$$

$$= r D_r\, 2\pi L$$

Now using the formula

$$Q_{enclosed} = \int_v \rho_v\,dv \qquad dv = r\,dr\,d\phi\,dz$$

$$= \int_v 10e^{-\tilde{r}}\,\tilde{r}\,d\tilde{r}\,d\phi\,dz$$

$$= \int_0^L \int_0^{2\pi} \int_0^r 10e^{-\tilde{r}}\,\tilde{r}\,d\tilde{r}\,d\phi\,dz$$

$$\int_0^r 10e^{-\tilde{r}}\,\tilde{r}\,d\tilde{r}\,d\phi\,dz$$

Using integration by parts.

$u = \tilde{r} \qquad v' = 10e^{-\tilde{r}}$

$u' = 1 \qquad v = -10e^{-\tilde{r}}$

$$u'v$$

$$\int_0^r 10e^{-\tilde{r}}\, d\tilde{r} = uv - \int \cancel{u'v}\, d\tilde{r}$$

$$= -\tilde{r}\,10e^{-\tilde{r}}\Big|_0^r - \int_0^r -10e^{-\tilde{r}}\, d\tilde{r}$$

$$= -\tilde{r}\,10e^{-\tilde{r}}\Big|_0^r + \int_0^r 10e^{-\tilde{r}}\, d\tilde{r}$$

$$= -r\,10e^{-r} + \Big[-10e^{-\tilde{r}}\Big]_0^r$$

$$= -r\,10e^{-r} + \Big[-10e^{-r} - (-10e^{0})\Big]$$

$$= -r\,10e^{-r} + \Big[-10e^{-r} + 10\Big]$$

$$\cancel{-r\,10e^{-r}}$$

$$= -r\,10e^{-r} - 10e^{-r} + 10$$

$$= 10e^{-r}(-r-1) + 10$$

∴ continuing the evaluation now

$$\int_V \rho_v\, dv = 10e^{-r}(-r-1)+10 \int_0^L \int_0^{2\pi} d\phi\, dz$$

$$= \left(10e^{-r}(-r-1)+10\right)(2\pi L)$$

∴

$$r D_r\,(\cancel{2\pi L}) = \left(10e^{-r}(-r-1)+10\right)(\cancel{2\pi L})$$

$$D_r = \frac{10e^{-r}}{r}(-r-1) + \frac{10}{r}$$

$$\vec{D} = \left(\frac{10e^{-r}}{r}(-r-1) + \frac{10}{r}\right)\hat{r}$$

Question 3 Continued

$$\frac{1}{r}\frac{\partial}{\partial r}\left(D_r r\right) = 10e^{-r}$$

$$\frac{\partial}{\partial r}\left(D_r r\right) = r10e^{-r}$$

$$\int_0^r \frac{\partial}{\partial r}(D_r r)\,dr = \int_0^r \tilde{r}10e^{-\tilde{r}}\,d\tilde{r}$$

$$D_r r = \int_0^r \tilde{r}\,10e^{-\tilde{r}}\,d\tilde{r}$$

Using integration by parts.

$$D_r r = -\tilde{r}10e^{-\tilde{r}}\Big|_0^r - \int_0^r -10e^{-\tilde{r}}\,d\tilde{r}$$

$$D_r r = -r10e^{-r} + \left[-10e^{-\tilde{r}}\right]_0^r$$

$$D_r r = -r10e^{-r} + \left[-10e^{-r} - (-10e^{0})\right]$$

$$D_r r = -r10e^{-r} + \left[-10e^{-r} + 10\right]$$

$$D_r r = -r10e^{-r} - 10e^{-r} + 10$$

$$D_r = \frac{10e^{-r}(-r-1)+10}{r}$$

$\frac{40}{40}$